

Advanced techniques in applied economics

Lecture 2: Gaussian and non-paranormal graphical models

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From conditional independence to networks

What did we learn last time?

Last time we introduced conditional independence. In particular, for Gaussian data,

$$X_i \perp\!\!\!\perp X_j \mid X_{\setminus ij} \iff K_{ij} = 0,$$

where $K = \Sigma^{-1}$ is the precision matrix.

Today we encode these conditional independence statements by **graphs**, and then learn those graphs from data.

Why might economists care about undirected graphs?

Undirected graphical models can be useful when we want a parsimonious summary of **conditional dependence structure** in high dimensions.

Examples from economics and finance

- Which asset returns remain associated after controlling for the rest?
- Which macro indicators still move together once common patterns are accounted for?
- Which sectors appear conditionally linked in stress periods?

Why this can be useful

- It reduces a very complex dependence structure to a sparse network.
- It helps distinguish marginal comovement from conditional association.
- It can serve as an exploratory map before moving to causal models.

Recall: In the Gaussian case covariance is marginal, precision is conditional.

Roadmap for today

1. Undirected graphs and conditional independence
2. Gaussian graphical models
3. Likelihood and estimation under graph constraints
4. High-dimensional learning and the graphical lasso
5. Robustness: non-paranormal graphical models
6. Application and interpretation

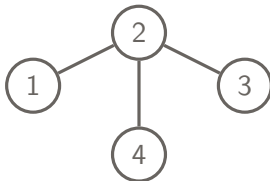
Part 1: Undirected graphical models

Undirected graphical models

Let $V = \{1, \dots, m\}$ index variables $X_1, \dots, X_m$. An **undirected graph** $G = (V, E)$ has

- one vertex for each variable,
- an edge $i - j$ if the model does *not* impose $X_i \perp\!\!\!\perp X_j \mid X_{V \setminus \{i,j\}}$.

Example: Suppose we have four variables X_1, X_2, X_3, X_4 and the graph

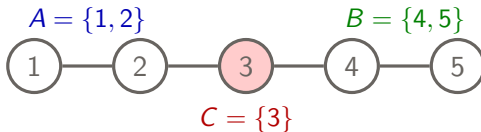


How should we read it?

The missing edge between 1 and 3 corresponds to $X_1 \perp\!\!\!\perp X_3 \mid X_2, X_4$. Likewise, $X_1 \perp\!\!\!\perp X_4 \mid X_2, X_3$.

Global Markov property

Let A , B , and C be disjoint subsets of vertices. If every path from A to B passes through C , then C **separates** A and B .



The graph encodes more CIs than just the pairwise: $X_i \perp\!\!\!\perp X_j \mid X_{V \setminus \{i,j\}}$.

Global Markov property: If C separates A and B in G , then

$$X_A \perp\!\!\!\perp X_B \mid X_C.$$

In the graph above, $(X_1, X_2) \perp\!\!\!\perp (X_4, X_5) \mid X_3$.

Clearly Global Markov Properties imply the Pairwise Markov Properties.

Hammersley–Clifford theorem

Graph factorization: If $C_1, \dots, C_r$ are the cliques of G , then

$$f(x) \propto \prod_{k=1}^r \psi_{C_k}(x_{C_k})$$

for suitable clique potentials ψ_{C_k} .

Hammersley-Clifford Theorem

For a **positive** distribution (full support), the following are equivalent:

1. the distribution is pairwise Markov with respect to the graph G ,
2. the distribution is global Markov with respect to the graph G ,
3. the density factorizes over the cliques of G .

Note that Gaussian density has full support.

Hammersley–Clifford connects graph separation with probabilistic factorization.

A tiny factorization example

Consider the chain graph



Its cliques are $\{1, 2\}$ and $\{2, 3\}$, so by Hammersley–Clifford a positive graphical model over this graph has density

$$f(x_1, x_2, x_3) \propto \psi_{12}(x_1, x_2) \psi_{23}(x_2, x_3).$$

Why this implies conditional independence

Fixing x_2 , we get

$$f(x_1, x_3 \mid x_2) = \frac{\psi_{12}(x_1, x_2)}{\sum_{x_1} \psi_{12}(x_1, x_2)} \frac{\psi_{23}(x_2, x_3)}{\sum_{x_3} \psi_{23}(x_2, x_3)}.$$

The first factor depends only on x_1 , the second only on x_3 , so

$$f(x_1, x_3 \mid x_2) = f(x_1 \mid x_2) f(x_3 \mid x_2).$$

Hence $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

Part 2: Gaussian graphical models

Gaussian graphical models

Let $X \sim \mathcal{N}_m(\mu, \Sigma)$ and let $K = \Sigma^{-1}$. Given an undirected graph $G = (V, E)$, the corresponding **Gaussian graphical model** is

$$\mathcal{M}(G) = \{\mathcal{N}_m(\mu, \Sigma) : K_{ij} = 0 \text{ whenever } i \not\sim j\}.$$

Key equivalence

For Gaussian data,

$$i \not\sim j \iff K_{ij} = 0 \iff X_i \perp\!\!\!\perp X_j \mid X_{V \setminus \{i,j\}}.$$

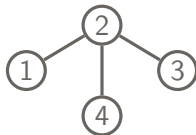
In Gaussian models, learning the graph is the same as learning the zero pattern of the precision matrix.

A four-variable example

Suppose

$$K = \begin{pmatrix} * & * & 0 & 0 \\ * & * & * & * \\ 0 & * & * & 0 \\ 0 & * & 0 & * \end{pmatrix}.$$

Then the corresponding graph is



The zeros $K_{13} = K_{14} = K_{34} = 0$ mean: $X_1 \perp\!\!\!\perp X_3 \mid X_{\{2,4\}}$, $X_1 \perp\!\!\!\perp X_4 \mid X_{\{2,3\}}$, $X_3 \perp\!\!\!\perp X_4 \mid X_{\{1,2\}}$.

Hammersley-Clifford gives, e.g., $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

The covariance matrix describes marginal comovement, while the precision matrix describes what remains after controlling for the other variables.

Part 3: Likelihood and estimation

The Gaussian likelihood

Suppose $X^{(1)}, \dots, X^{(n)} \stackrel{iid}{\sim} \mathcal{N}_m(\mu, \Sigma)$ and let

$$S = \frac{1}{n} \sum_{k=1}^n (X^{(k)} - \bar{X})(X^{(k)} - \bar{X})^\top$$

be the sample covariance matrix.

Profile log-likelihood in $K = \Sigma^{-1}$

Up to an additive constant,

$$\ell(K) = \frac{n}{2} (\log \det K - \text{tr}(SK)).$$

Unconstrained MLE

If there are no graph constraints and S is invertible, the MLE is

$$\hat{K} = S^{-1}.$$

Estimation under a fixed graph

If the graph G is known, we maximize the Gaussian likelihood subject to the zero constraints

$$K_{ij} = 0 \quad \text{for all } i \not\sim j.$$

Constrained MLE: maximize over $K \in \mathbb{S}_+^m$ such that $K_{ij} = 0$ if $i \not\sim j$

$$\hat{K}_G = \arg \max \left\{ \log \det K - \text{tr}(SK) \right\}.$$

- There is a unique optimizer and there are many good algorithms to find the optimum.

Interpretation

The most likely distribution whose direct-dependence pattern agrees with the graph.

What is hard?

Even for Gaussian data, two different problems must be separated.

1. **Estimation:** if the graph is known, estimate K under the zero constraints.
2. **Structure learning:** if the graph is unknown, search over many possible zero patterns.

Combinatorial explosion

The number of undirected graphs on m vertices is

$$2^{\binom{m}{2}}.$$

So exhaustive search is impossible even for moderate m .

The real challenge is to learn the sparsity pattern.

Why naive covariance inversion fails

When the dimension m is large relative to the sample size n :

- S is noisy
- S^{-1} is unstable
- if $m > n$, then S is singular.

Not the first time you encounter this problem

This is the same basic problem as in high-dimensional regression: too many unknown parameters, not enough data.

Regression

Estimate $\beta \in \mathbb{R}^p$ (p large) in

$$Y = X\beta + \varepsilon.$$

Solution: impose sparsity with LASSO.

Graph learning

Estimate the precision matrix $K = \Sigma^{-1} \in \mathbb{S}_+^m$.

Here m is large.

Solution: impose sparsity on K .

Graphical LASSO is the precision-matrix analogue of LASSO regression.

A quick concentration experiment

Even in a simple Gaussian model, estimating a large covariance matrix is difficult.

```
library(MASS)
set.seed(1)
m <- 100
n <- 80
Sigma <- diag(m)
X <- mvrnorm(n, mu = rep(0,m), Sigma = Sigma)
S <- cov(X)
range(eigen(S)$values)

det(S) # numerically unstable or zero when m >= n
```

Problem

The sample covariance matrix may be poorly behaved exactly in the regimes where network methods are most needed.

Graphical lasso

A standard approach is to add an ℓ_1 penalty to encourage sparsity:

$$\hat{K}_\lambda = \arg \max_{K \succ 0} \left\{ \log \det K - \text{tr}(SK) - \lambda \sum_{i \neq j} |K_{ij}| \right\}.$$

The role of the penalty parameter λ

The penalty shrinks many off-diagonal entries toward zero, producing a sparse estimated graph.

Why the ℓ_1 penalty helps?

Economic reason

A sparse graph encodes the belief that most pairs of variables are not directly linked once the rest is controlled for.

Statistical reason

It reduces variance and stabilizes estimation when m is large.

Interpretive reason

It produces sparse networks that can actually be read and discussed.

Choosing the penalty parameter

In practice, λ must be chosen from the data. Common approaches include:

- cross-validation,
- information criteria such as BIC or EBIC,
- stability-based methods.

Practical lesson

There is no single “correct” graph. What we estimate depends on the statistical criterion and the amount of regularization.

EBIC-based choices are convenient and are implemented in packages such as `qgraph` and `huge`.

A first Gaussian graph in R

Example data

The dataset `stockdata` contains stock-return data. Here we take the first 40 variables, standardize them, and estimate a sparse Gaussian graphical model.

```
library(huge)
library(qgraph)
data(stockdata)

x <- scale(stockdata$data[,1:40])

out <- huge(x, method = "glasso")
plot(out)
```

```
# pick one graph along the
# regularization path
Khat <- out$icov[[10]]
Khat <- (Khat + t(Khat))/2

qgraph(-cov2cor(Khat),
       layout = "spring")
```

What is the path?

`huge(..., method = "glasso")` fits the graphical LASSO for many values of the penalty parameter λ . This gives a whole **regularization path**, from denser graphs to sparser ones. The command `out$icov[[10]]` picks one particular estimate along that path.

Part 5: Beyond Gaussianity

Why Gaussianity is often too optimistic

Economic data are rarely exactly Gaussian.

- asset returns are heavy-tailed,
- macro and firm-level variables may be skewed,
- relationships are often nonlinear,
- monotone rescalings can substantially change Pearson correlations.

We would like a model that keeps the Gaussian graphical machinery, but is less sensitive to non-Gaussian marginals.

Gaussian copulas and the non-paranormal model

A flexible alternative is to assume that the observed variables are monotone transformations of latent Gaussian variables.

Non-paranormal model

We say that $X = (X_1, \dots, X_m)$ is **non-paranormal** if there exist strictly monotone functions $f_1, \dots, f_m$ such that

$$(f_1(X_1), \dots, f_m(X_m)) \sim \mathcal{N}_m(\mu, \Sigma).$$

Non-paranormal models relax Gaussian marginals without abandoning Gaussian graphical structure.

Why monotone transformations preserve the graph

Suppose

$$Y_i = f_i(X_i), \quad i = 1, \dots, m,$$

where each f_i is strictly monotone.

Key fact

Such transformations preserve conditional independence:

$$X_i \perp\!\!\!\perp X_j \mid X_{\setminus ij} \iff Y_i \perp\!\!\!\perp Y_j \mid Y_{\setminus ij}.$$

Monotone transformations may change means, variances, and Pearson correlations, but they do not change the underlying conditional independence graph.

So if the data become Gaussian after unknown monotone transformations, we can still hope to recover the same graph from the observed variables.

Kendall's tau and monotone invariance

For two real-valued variables X_i and X_j , Kendall's tau is defined by

$$\tau_{ij} = \mathbb{P}((X_i - \tilde{X}_i)(X_j - \tilde{X}_j) > 0) - \mathbb{P}((X_i - \tilde{X}_i)(X_j - \tilde{X}_j) < 0),$$

where $(\tilde{X}_i, \tilde{X}_j)$ is an independent copy of (X_i, X_j) .

Kendall's tau is a rank-based measure of dependence. It compares how often two observations move in the same order (concordant pairs) versus opposite order (discordant pairs).

Crucial property

Kendall's tau is invariant under strictly monotone transformations:

$$\tau(f_i(X_i), f_j(X_j)) = \tau(X_i, X_j).$$

Kendall's tau as a bridge to latent Gaussian correlation

If (X_i, X_j) is bivariate Gaussian with correlation Σ_{ij} , then

$$\tau_{ij} = \frac{2}{\pi} \arcsin(\Sigma_{ij}), \quad \text{equivalently} \quad \Sigma_{ij} = \sin\left(\frac{\pi}{2} \tau_{ij}\right).$$

Why this is powerful

Suppose we observe: $Y_i = f_i(X_i), \quad i = 1, \dots, m,$

where the latent vector X is Gaussian and the functions f_i are unknown but monotone. Then

$$\tau(Y_i, Y_j) = \tau(X_i, X_j),$$

so we can recover the latent Gaussian correlation by

$$\Sigma_{ij} = \sin\left(\frac{\pi}{2} \tau(Y_i, Y_j)\right).$$

This gives a way to estimate the latent Gaussian correlation matrix from the observed data **without knowing the transformations f_i .**

Estimating a non-paranormal graph

A simple plug-in strategy is:

1. compute the matrix of sample Kendall's taus $\hat{\tau}$,
2. convert it into a latent correlation estimate

$$\hat{S}_{ij}^{\tau} = \sin\left(\frac{\pi}{2}\hat{\tau}_{ij}\right),$$

3. apply a standard Gaussian graph-learning method to $\hat{S}^{\tau}$.

Estimate the latent Gaussian correlation first, then estimate the sparse graph.

Non-paranormal estimation in R

```
library(huge)
data(stockdata)

x <- scale(stockdata$data[,1:40])

# rank-based transformation
x.npn <- huge.npn(x,
                  npn.func = "skeptical")
```

```
out.npn <- huge(x.npn,
                method = "glasso")

Khat <- out.npn$icov[[10]]
Khat <- (Khat + t(Khat))/2

qgraph(-cov2cor(Khat),
       layout = "spring")
```

Interpretation

This estimates a sparse graph after a rank-based non-paranormal correction.

Gaussian vs non-paranormal networks

Pure Gaussian approach

Uses the sample covariance or correlation matrix directly. Fast and elegant, but sensitive to heavy tails and marginal distortions.

Non-paranormal approach

Uses rank information to recover a latent Gaussian correlation structure. More robust when the marginals are non-Gaussian.

For economic data, robustness to monotone distortions is often worth the extra step.

Part 6: Interpreting estimated networks

What an estimated graph can tell us

A sparse graph can help us:

- identify clusters of directly related variables,
- separate broad comovement from direct links,
- generate hypotheses about sectors, channels, and transmission mechanisms,
- reduce a high-dimensional system to a manageable network summary.

Example: stock returns

We may see sectoral clusters, central firms, and weak cross-sector connections after conditioning on the rest of the market.

What an estimated graph cannot tell us

A graph learned from observational data does **not** automatically provide:

- causal direction,
- intervention effects,
- economic mechanisms by itself,

Important caution

An undirected edge means “conditional association remains.” It does **not** mean “ i causes j .”

This is why our next block of lectures will move from **undirected structure** to **directed causal structure**.

Take-away

- Undirected graphs encode conditional independence statements.
- In Gaussian graphical models, missing edges correspond to zeros in the precision matrix.
- The Hammersley–Clifford theorem links graph separation with factorization.
- In high dimensions, sparse estimation requires regularization such as graphical lasso.
- Non-paranormal methods make the same ideas more robust for heavy-tailed and non-Gaussian data.

Main lesson of today: conditional independence becomes a network, and sparsity makes that network learnable.

Looking ahead

Next lecture

We move to directed graphs. Instead of symmetric conditional dependence, we will study asymmetry, causal ordering, and interventions.

Coming next

- DAGs and d-separation,
- Markov equivalence,
- structural equation models,
- soft interventions and policy interpretation.